WORKED SOLUTION

STEP II 2004, Question 6

Pure • Vector Decomposition

> **STEP II 2004, Question 6**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/04-s2-q6

The key idea

Splitting a vector into a part along $\mathbf{a}$ and a part perpendicular to $\mathbf{a}$ is just projection: since $\mathbf{a}$ is a unit vector, the component of any vector $\mathbf{v}$ along it is $(\mathbf{v}\cdot\mathbf{a})\,\mathbf{a}$. The second part is the same idea one dimension up. The vectors $\mathbf{a}$ and $\mathbf{q}$ are perpendicular directions in Π, and $\mathbf{R}$ supplies the third, perpendicular direction. So we can either project onto each direction in turn or write down unknown multiples and impose the perpendicularity conditions with dot products. We do the second and use the first as a check.

Solution

The vectors p and q

Since $\mathbf{p}$ is parallel to $\mathbf{a}$, write $\mathbf{p}=\lambda\mathbf{a}$. Then

$$\mathbf{q}=\mathbf{a}+\mathbf{b}-\lambda\mathbf{a}=(1-\lambda)\mathbf{a}+\mathbf{b},$$

which lies in Π automatically, because $\mathbf{a}$ and $\mathbf{b}$ do. For $\mathbf{q}$ to be perpendicular to $\mathbf{a}$ we need

$$0=\mathbf{q}\cdot\mathbf{a}=(1-\lambda)\,|\mathbf{a}|^2+\mathbf{a}\cdot\mathbf{b}=(1-\lambda)+3,$$

using $|\mathbf{a}|=1$ and $\mathbf{a}\cdot\mathbf{b}=3$. So $\lambda=4$ and

$$\boxed{\mathbf{p}=4\mathbf{a},\qquad \mathbf{q}=\mathbf{b}-3\mathbf{a}.}$$

Indeed $\mathbf{q}\cdot\mathbf{a}=3-3=0$, and $\mathbf{p}+\mathbf{q}=\mathbf{a}+\mathbf{b}$. (Equivalently, $\mathbf{p}$ is the projection $((\mathbf{a}+\mathbf{b})\cdot\mathbf{a})\mathbf{a}=4\mathbf{a}$.)

The vectors P, Q and R

First, a few facts we will need. With $|\mathbf{b}|=5$,

$$|\mathbf{q}|^2=|\mathbf{b}|^2-6\,\mathbf{a}\cdot\mathbf{b}+9|\mathbf{a}|^2=25-18+9=16,$$

so $|\mathbf{q}| = 4$. In particular $\mathbf{q} \neq \mathbf{0}$, so $\mathbf{b}$ is not parallel to $\mathbf{a}$, and $\mathbf{a}$ and $\mathbf{b}$ span the plane Π.

Write $\mathbf{P} = \alpha\mathbf{a}$ and $\mathbf{Q} = \beta\mathbf{q} = \beta(\mathbf{b} - 3\mathbf{a})$. Then $\mathbf{R}$ is forced to be

$$\mathbf{R} = \mathbf{a} + \mathbf{b} + \mathbf{c} - \alpha\mathbf{a} - \beta(\mathbf{b} - 3\mathbf{a}) = \mathbf{c} + u\,\mathbf{a} + v\,\mathbf{b}, \qquad \text{where } u = 1 - \alpha + 3\beta,\ v = 1 - \beta.$$

Ask yourself How do we turn "$\mathbf{R}$ is perpendicular to Π" into equations?

A vector is perpendicular to a plane exactly when it is perpendicular to two non-parallel vectors in that plane. We have just seen that $\mathbf{a}$ and $\mathbf{b}$ are such a pair. So the condition is $\mathbf{R} \cdot \mathbf{a} = 0$ and $\mathbf{R} \cdot \mathbf{b} = 0$. Using $\mathbf{a} \cdot \mathbf{c} = -2$, $\mathbf{b} \cdot \mathbf{c} = 2$, $\mathbf{a} \cdot \mathbf{b} = 3$, $|\mathbf{a}|^2 = 1$ and $|\mathbf{b}|^2 = 25$:

$$\mathbf{R} \cdot \mathbf{a} = -2 + u + 3v = 0,$$
$$\mathbf{R} \cdot \mathbf{b} = 2 + 3u + 25v = 0.$$

The first gives $u = 2 - 3v$. Substituting into the second gives $8 + 16v = 0$, so

$$v = -\tfrac{1}{2}, \qquad u = \tfrac{7}{2}.$$

Then $\beta = 1 - v = \tfrac{3}{2}$ and $\alpha = 1 + 3\beta - u = 1 + \tfrac{9}{2} - \tfrac{7}{2} = 2$. Therefore

$$\boxed{\mathbf{P} = 2\mathbf{a}, \qquad \mathbf{Q} = \tfrac{3}{2}(\mathbf{b} - 3\mathbf{a}), \qquad \mathbf{R} = \mathbf{c} + \tfrac{7}{2}\mathbf{a} - \tfrac{1}{2}\mathbf{b}.}$$

Adding these, the coefficient of $\mathbf{a}$ is $2 - \tfrac{9}{2} + \tfrac{7}{2} = 1$, the coefficient of $\mathbf{b}$ is $\tfrac{3}{2} - \tfrac{1}{2} = 1$ and the coefficient of $\mathbf{c}$ is 1, so $\mathbf{P} + \mathbf{Q} + \mathbf{R} = \mathbf{a} + \mathbf{b} + \mathbf{c}$ as required.

Finally, $\mathbf{R} \neq \mathbf{0}$: if it were zero, $\mathbf{c} = \tfrac{1}{2}\mathbf{b} - \tfrac{7}{2}\mathbf{a}$ would lie in Π, contrary to the hypothesis that $\mathbf{c}$ is not parallel to Π. That hypothesis is what makes $\mathbf{R}$ a genuine normal vector.

A second view: projecting onto perpendicular axes

The unit vectors $\mathbf{a}$, $\hat{\mathbf{q}} = \tfrac{1}{4}(\mathbf{b} - 3\mathbf{a})$ and a unit normal $\mathbf{n}$ to Π form an orthonormal set. So $\mathbf{P}$ and $\mathbf{Q}$ are the projections of $\mathbf{v} = \mathbf{a} + \mathbf{b} + \mathbf{c}$ onto the first two:

$$\mathbf{v} \cdot \mathbf{a} = 1 + 3 - 2 = 2, \qquad \text{so } \mathbf{P} = 2\mathbf{a};$$
$$\mathbf{v} \cdot \hat{\mathbf{q}} = \tfrac{1}{4}(\mathbf{a} \cdot \mathbf{q} + \mathbf{b} \cdot \mathbf{q} + \mathbf{c} \cdot \mathbf{q}) = \tfrac{1}{4}(0 + 16 + 8) = 6, \qquad \text{so } \mathbf{Q} = 6\hat{\mathbf{q}} = \tfrac{3}{2}(\mathbf{b} - 3\mathbf{a}),$$

using $\mathbf{b} \cdot \mathbf{q} = 25 - 9 = 16$ and $\mathbf{c} \cdot \mathbf{q} = 2 + 6 = 8$. What is left over, $\mathbf{R}$, is the component of $\mathbf{c}$ normal to Π. This agrees with the answer above.

Checks

- Directly: $\mathbf{R} \cdot \mathbf{a} = -2 + \tfrac{7}{2} - \tfrac{3}{2} = 0$ and $\mathbf{R} \cdot \mathbf{b} = 2 + \tfrac{21}{2} - \tfrac{25}{2} = 0$.
- A concrete case: take $\mathbf{a} = \mathbf{i}$, $\mathbf{b} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{c} = -2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$, which satisfy all the given conditions. Then $\mathbf{q} = 4\mathbf{j}$, $\mathbf{P} = 2\mathbf{i}$, $\mathbf{Q} = 6\mathbf{j}$ and $\mathbf{R} = \mathbf{k}$, and these add up to $\mathbf{a} + \mathbf{b} + \mathbf{c} = 2\mathbf{i} + 6\mathbf{j} + \mathbf{k}$.
- We also tested the formulae on 2000 randomly oriented configurations satisfying the conditions. Every time, $\mathbf{q} \cdot \mathbf{a} = 0$, $|\mathbf{q}| = 4$, $\mathbf{R}$ was normal to Π and the three vectors summed to $\mathbf{a} + \mathbf{b} + \mathbf{c}$, all to 10^{-9}.

Common slips

- Taking $\mathbf{p} = (\mathbf{a}\cdot\mathbf{b})\mathbf{a} = 3\mathbf{a}$, which is the projection of $\mathbf{b}$ alone. We need the projection of $\mathbf{a}+\mathbf{b}$, which is $4\mathbf{a}$.
- Imposing only $\mathbf{R}\cdot\mathbf{a} = 0$. Perpendicularity to a plane needs two independent conditions.
- Assuming $\mathbf{P} = \mathbf{p}$ and $\mathbf{Q} = \mathbf{q}$. Adding $\mathbf{c}$ changes the in-plane components too, because $\mathbf{c}$ itself has a component in Π.

Beyond STEP

This is the Gram–Schmidt process in miniature. From $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ we built mutually perpendicular directions $\mathbf{a}$, $\mathbf{b}-3\mathbf{a}$ and $\mathbf{c}+\frac{7}{2}\mathbf{a}-\frac{1}{2}\mathbf{b}$, and then decomposed a vector along them. The same procedure works in any number of dimensions. It underlies orthogonal projection, least-squares fitting and the QR factorisation of a matrix. Strang's *Introduction to Linear Algebra* develops all of these.

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